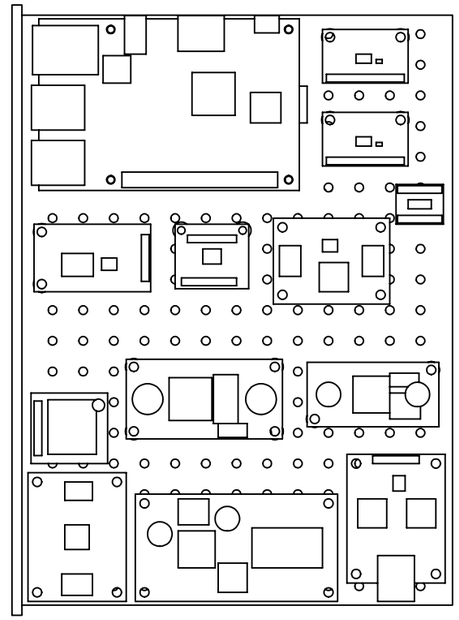
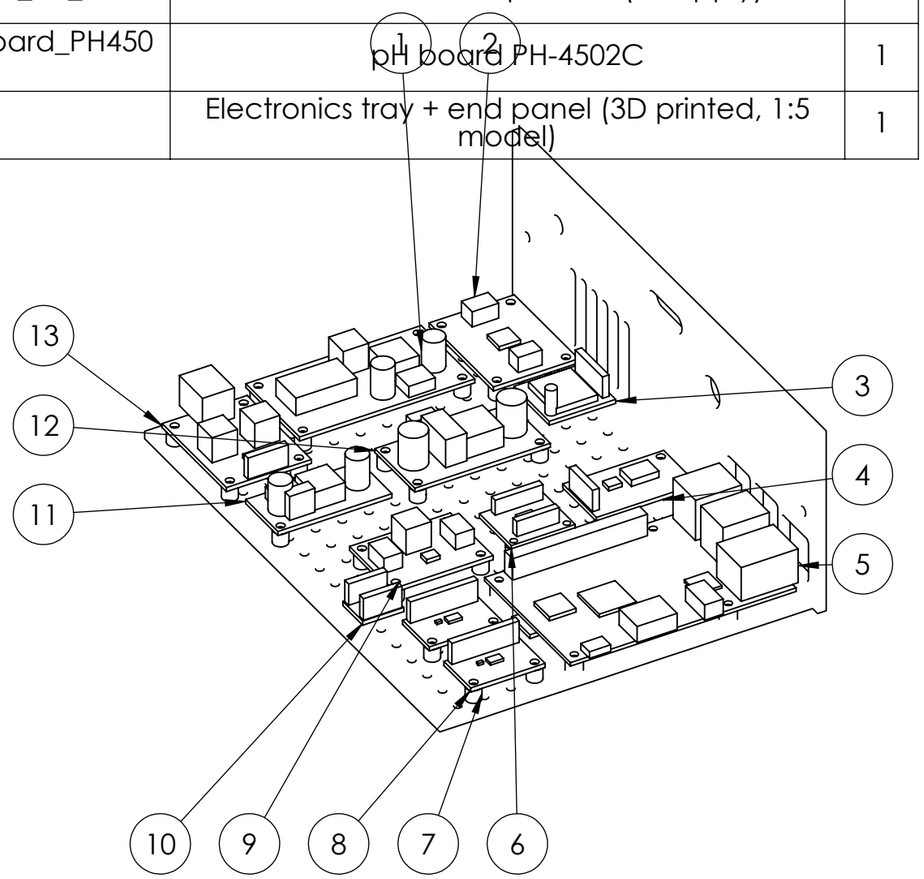
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	E06_Buck_B_SIM800L_4V	Buck B: LM2596 with voltmeter (SIM800L 4.0 V)	1
2	E10_TDS_board_SEN0244	TDS Meter V1.0 board	1
3	E04_SIM800L_GSM	SIM800L GSM/GPRS module	1
4	E03_DS3231_RTC	DS3231 real-time clock (ZS-042)	1
5	E01_Raspberry_Pi_3B_plus	Raspberry Pi 3 Model B+	1
6	E08_MX1508_pump_driver	MX1508 "mini L298N" pump driver	1
7	E13_Standoff_M3x5_nylon	Nylon standoff M3 × 5	30
8	E02_ADS1115_ADC	ADS1115 16-bit ADC module	2
9	E11_Turbidity_adapter	Turbidity sensor adapter (SEN0189)	1
10	E12_Level_converter	4-channel logic level converter	1
11	E07_Buck_C_LM2596S_5V	Buck C: LM2596S (5 V sensor rail)	1
12	E05_Buck_A_5V_5A	Buck A: 5 V 5 A step-down (Pi supply)	1
13	E09_pH_board_PH4502C	pH board PH-4502C	1
14	LD-S08	Electronics tray + end panel (3D printed, 1:5 model)	1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE			TITLE: Electronics bay, 1:5 LIVE model	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. AS-LIVE-ELEC-01	A3
				WEIGHT:		SCALE:1:2	SHEET 1 OF 2

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RASPBERRY PI 3B+ PIN MAP (BCM)

Pi pin	GPIO	Connects to	Notes
1	3.3 V	ADS1115 x2, DS3231, DHT11, DS18B20 pull-up, rain + LDR modules	about 20 mA total
2, 4	5 V	Pi supply from Buck A	external 3 A fuse
3 / 5	GPIO2 / 3	I2C SDA / SCL: ADS1115 0x48, 0x49, DS3231 0x68	module pull-ups
7	GPIO4	DS18B20 data (1-Wire)	4.7 k to 3.3 V
8	GPIO14 TXD	SIM800L RXD via 1 k / 2 k divider	3.3 V to 2.2 V
10	GPIO15 RXD	SIM800L TXD	2.8 V level, read OK
11	GPIO17	DHT11 data	dtoverlay dht11
12	GPIO18	SIM800L RST	
13	GPIO27	Rain module DO	
15	GPIO22	Rain module power switch	on only while reading
16	GPIO23	MX1508 IN1 (pump)	IN2 tied low
29	GPIO5	LDR module DO	
31	GPIO6	pH board 5 V high-side switch	never with GPIO13
33	GPIO13	TDS board high-side switch	
35	GPIO19	MQ-3 heater MOSFET	about 180 mA
36	GPIO16	ACK / reset button	internal pull-up
37	GPIO26	Status LED / beacon	via transistor
6, 9, 14	GND	Star ground at ADS1115 #1	

ADC CHANNELS

ADC	Ch	Signal	Raw range	Conditioning	At the ADC
#1 0x48	A0	pH Po	0-5 V	12 k / 20 k, 1 k + 100 nF	0-3.1 V
#1	A1	TDS	0-2.3 V	1 k + 100 nF	0-2.3 V
#1	A2	Turbidity	0-4.5 V	12 k / 20 k, 1 k + 100 nF	0-2.8 V
#1	A3	MQ-3 AO	0-5 V	12 k / 20 k, 1 k + 100 nF	0-3.1 V
#2 0x49	A0	LDR AO	0-3.3 V	100 nF	0-3.3 V
#2	A1	Rain AO	0-3.3 V	100 nF	0-3.3 V
#2	A2	Battery voltage	0-14.6 V	100 k / 22 k	0-2.6 V
#2	A3	Spare			

POWER RAILS

Rail	From	Feeds	Protection
12 V in	12 V 3 A adapter (bench) or LiFePO4 12.8 V (field)	Buck A, B, C; pump	3 A fuse, reverse polarity, TVS
5.0 V	Buck A (5 V 5 A)	Raspberry Pi 3B+	external fuse on the 5 V pins
4.0 V	Buck B (LM2596 with voltmeter)	SIM800L VBAT	1000 uF low-ESR + 100 nF + 5.1 V Zener
5.0 V	Buck C (LM2596S)	pH, TDS, turbidity, MQ-3 boards	switched per sensor
3.3 V	Raspberry Pi pin 1	ADCs, RTC, DHT11, DS18B20, rain, LDR	about 20 mA

UNLESS OTHERWISE SPECIFIED:
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SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE:

Electronics bay, 1:5 LIVE model

DWG NO. AS-LIVE-ELEC-01

A3

SCALE:1:1

SHEET 2 OF 2

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